

Literature Summary Report

1. Image Steganography Algorithm Recognition Based on Big Data Matching with Improved ResNet50

Developed a deep learning model for detecting hidden information inside digital images using an improved ResNet50 architecture. The model combines big data matching techniques with convolutional neural networks to identify steganographic patterns that are not visible to the human eye. The system focuses on analyzing image features and detecting abnormal embedding artifacts.

The experiments were conducted using image datasets containing steganographic and normal images. The improved ResNet50 achieved high detection accuracy and better feature extraction compared to traditional CNN models.

The novelty of this paper lies in enhancing ResNet50 with improved feature matching techniques for better steganography detection. This work suggests that deep CNN-based feature extraction can effectively identify hidden malicious content in images. However, the limitation is that it detects hidden patterns but does not extract or classify URLs embedded inside images.

2. A Differential Evolution-Based Hyperparameter Optimization Framework for Image Steganalysis

Proposed a framework that uses Differential Evolution (DE) optimization to automatically tune hyperparameters of machine learning models used in image steganalysis. Instead of manually selecting parameters, the optimization algorithm searches for the best combination to improve detection performance.

The model was tested on image datasets containing steganographic and normal images. The optimized framework achieved improved accuracy and robustness compared to non-optimized models.

The novelty of this paper lies in integrating evolutionary optimization with steganalysis models, improving performance without manual tuning. This paper teaches that intelligent optimization techniques can significantly improve detection systems. However, it focuses only on hidden data detection and does not address malicious URL extraction from images.

3. Malicious URL Detection with Explainable Machine Learning Techniques

Developed a malicious URL detection system using explainable machine learning models such as Logistic Regression, Decision Tree, and Random Forest. The goal was not only to achieve high accuracy but also to provide transparency in predictions.

The experiments were conducted on the Kaggle malicious URL dataset. Among all models, Random Forest achieved the highest accuracy of 94.2%, with good precision and recall values.

The novelty of this paper lies in combining strong classification performance with explainability. The system helps understand which features contribute to malicious URL detection. However, the limitation is that it works only on textual URLs and does not analyze URLs hidden inside images.

4. An Integrated CSPPC–BiLSTM Framework for Malicious URL Detection

Proposed a deep learning framework combining Character Embedding, BiLSTM, Attention mechanism (CBAM), and Spatial Pyramid Pooling for malicious URL detection. The model captures both local and global patterns in long URL sequences.

The system was evaluated on the Gramembedding dataset and Mendeley AK Singh phishing dataset. It achieved very high accuracy (98.73% for balanced dataset and 97.26% for imbalanced dataset) along with high F1-score.

The novelty of this paper lies in integrating advanced deep learning layers to handle long and imbalanced URL datasets effectively. This paper demonstrates that deep sequence models significantly improve textual URL detection. However, it does not handle image-based or visually hidden URLs.

5. MalJPEG: Machine Learning Based Solution for the Detection of Malicious JPEG Images

Developed a system called MalJPEG that detects malicious content hidden inside JPEG image files. Instead of analyzing the visible image content, the model extracts structural features from JPEG file headers and metadata to detect malicious modifications.

The model was tested on large-scale real-world JPEG datasets. It achieved very high performance with AUC of 99.1%, True Positive Rate of 98.4%, and False Positive Rate of 1.6%.

The novelty of this paper lies in analyzing internal JPEG structure rather than visual features. It proves that file-level analysis can effectively detect malicious images. However, it does not extract or classify URLs embedded in images.

6. Cyber Threat Intelligence Sharing Scheme Based on Federated Learning for Network Intrusion Detection

Proposed a federated learning-based framework for network intrusion detection. The main objective was to allow multiple organizations to collaboratively train a detection model without sharing raw data, thus preserving privacy.

The model was evaluated on NF-UNSW-NB15-v2 and NF-BoT-IoT-v2 datasets and achieved very high accuracy (above 98%). Federated learning helped improve detection performance while maintaining data confidentiality.

The novelty of this paper lies in applying federated learning to cyber threat intelligence sharing. This improves security and privacy simultaneously. However, this approach focuses on network traffic analysis and is not related to image-based malicious URL detection.

7. An Intelligent Cyber Threat Detection: A Swarm-Optimized Machine Learning Approach

Proposed a cyber threat detection system that combines Harris Hawks Optimization (HHO) with Support Vector Machine (SVM). The optimization algorithm selects the best features and parameters to improve classification performance.

The system was tested on malicious URLs, spam emails, spam websites, and IoT intrusion datasets. It achieved very high accuracy, including 100% accuracy for malicious URL detection.

The novelty of this paper lies in combining swarm intelligence optimization with machine learning classifiers to boost performance. The optimization improves accuracy significantly. However, the limitation is that it works only on structured features and does not process image-based or OCR-extracted URLs.